



UNDERWATER ROBOTICS LOCOMOTION

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KEY DESIGN ISSUES

- Buoyancy & Pressure Resistance: Underwater robots must maintain neutral or adjustable buoyancy and withstand high hydrostatic pressures.
- Hydrodynamics: Efficient movement through water requires a streamlined design to reduce drag.

Propulsion:

- Propellers or thrusters for ROVs (Remotely Operated Vehicles)
- Fin-based propulsion for biomimetic AUVs (Autonomous Underwater Vehicles)
- Corrosion Resistance: Use of marine-grade materials (e.g., titanium, composites) to prevent rust and degradation.

ATTRIBUTES



- Endurance: Depends on battery capacity and energy efficiency.
- Maneuverability: Ability to change direction in 3D space (pitch, yaw, roll).
- Depth Range: Operational depth determined by pressure tolerance and mission requirements.
- Stability: Controlled via internal ballast systems or dynamic fins.

BASIC THEORETICAL MODEL



Kinematic Models:

- 6 Degrees of Freedom (DOF): Surge, sway, heave, roll, pitch, yaw.
- Newton-Euler equations to describe motion.

Dynamic Considerations:

- Water drag, lift forces, buoyant forces.
 - Nonlinearities due to fluid dynamics.
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PAYLOAD



- Cameras: Optical and low-light for inspection and exploration.
- Sonar Systems: For navigation, obstacle avoidance, and mapping (e.g., side-scan sonar).
- Manipulators: Robotic arms for underwater sampling, repair, or interaction.
- Sensors: Depth sensors, inertial navigation systems (INS), pressure transducers.

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